

ACIC 200/400/800 Tablet

ACIC 200 Tablet

Each tablet contains: Aciclovir 200mg. The tablet shall be white, round, biconvex with a score notch and a smooth surface. Diameter of the tablet is 12.1 ± 0.2 mm and height 4.5 ± 0.4 mm.

ACIC 400 Tablet

Each tablet contains: Aciclovir 400mg. The tablet shall be white, round, biconvex with a score notch and a smooth surface. Diameter of the tablet is 12.1 ± 0.2 mm and height 5.5 ± 0.4 mm.

ACIC 800 Tablet

Each tablet contains: Aciclovir 800mg. The tablet shall be white, oblong tablets with score notches on both sides and a smooth surface. Diameter of the tablet is 19.1 ± 0.2 mm and width 8.1 ± 0.2 mm and height 7.8 ± 0.5 mm.

Properties

Aciclovir inhibits viral replication by inhibiting the viral DNA synthesis. The aciclovir is phosphorylated by viral thymidine kinase. Aciclovir has 200 times greater affinity to herpes-virus-encoded thymidine kinase as compare to the mammalian thymidine kinase. The phosphorylation of aciclovir leads to formation of aciclovir monophosphate (aciclo-GMP). Aciclo-GMP is then catalysed by the normal cellular enzymes to form the aciclo-GDP & aciclo-GTP. Aciclo-GTP acts by competitive inhibition of viral DNA polymerase formation and the incorporation of aciclo-GTP into the viral DNA terminates the biosynthesis of the viral DNA strand.

Aciclovir is a synthetic purine nucleoside analog, 9-((2-hydroxyethoxy) methyl) guanine with a linear side chain substituted for the cyclic sugar of the guanosine molecule.

Aciclovir is active against the herpes viruses especially the herpes simplex type 1 and herpes simplex type 2. Aciclovir is less sensitive to the Varicella-zoster virus but is inhibited by concentrations that can be achieved clinically. Aciclovir has been proven to be 100 times more active than vidarabine and 10 times more active than idoxudine against herpes simplex type 1 in vitro. Aciclovir is selective against the viral infected cells and uninfected mammalian cells are not affected by aciclovir unless the concentrations exceed $70 \mu\text{g/ml}$.

Pharmacokinetics

The peak plasma concentration of aciclovir after intravenous doses of 5 mg/kg is approximately $10 \mu\text{g/ml}$. This concentration will decrease to a trough of $0.7 \mu\text{g/ml}$ after 8 hours. Aciclovir is partially absorbed (approximately 20%) after oral administration. Mean peak steady state concentration after oral dose of 200mg four hourly is approximately $3.1 \mu\text{mol}$ ($0.7 \mu\text{g/ml}$) and the mean trough $1.8 \mu\text{mol}$ ($0.4 \mu\text{g/ml}$). Mean peak steady state concentration after oral dose of 400mg and 800mg four hourly give a level of $5.3 \mu\text{mol}$ ($1.2 \mu\text{g/ml}$) and $8 \mu\text{mol}$ ($1.8 \mu\text{g/ml}$) corresponding trough of $2.7 \mu\text{mol}$ ($0.6 \mu\text{g/ml}$) and $4 \mu\text{mol}$ ($0.9 \mu\text{g/ml}$). The elimination of acyclovir is mainly by glomerular filtration and tubular secretion. The beta-phase half-life is about 2.5-2.9 hours for adult with normal kidney function. Approximately 15% of administered dose is excreted in urine as the inactive metabolite, 9-carboxymethoxymethylguanine. Aciclovir is well distributed and the volume of distribution corresponds to total body water; the concentration in cerebrospinal fluid and aqueous humor are only one third to one half of those in plasma.

Indication

Herpes simplex, particularly herpes genitalis infections of the skin and mucosa membranes (including both initial and recurrent genital herpes). Suppression of recurrent herpes simplex infections in immunocompetent patients. Prophylaxis of herpes simplex infections in immune compromised patients. Treatment of Varicella (chickenpox) and Herpes zoster (shingles) infections.

Adverse Reactions

• Blood and lymphatic system disorders

Very rare: Anaemia, leukopenia, thrombocytopenia

• Immune system disorders

Rare: Anaphylaxis

• Psychiatric and nervous system disorders

Common: Headache, dizziness

Very rare: Agitation, confusion, tremor, ataxia, dysarthria, hallucinations, psychotic symptoms, convulsions, somnolence, encephalopathy and coma.

• Respiratory, thoracic and mediastinal disorders

Rare: Dyspnoea

• Gastrointestinal disorders

Common: Nausea, vomiting, diarrhoea, abdominal pains

• Hepato-biliary disorders

Rare: Reversible rises in bilirubin and liver related enzymes

Very rare: Hepatitis, jaundice

• Skin and subcutaneous tissue disorders

Common: Pruritus, rashes (including photosensitivity)

Uncommon: Urticaria, accelerated diffuse hair loss

Rare: Angioedema

• Renal and urinary disorders

Rare: Increases in blood urea and creatinine

Very rare: Acute renal failure, renal pain

• General disorders and administration site conditions

Common: Fatigue, fever

Precaution/warnings

Adequate hydration should be maintained in patients given high oral doses of aciclovir.

The risk of renal impairment is increased by use with other nephrotoxic active substances.

Use in patients with renal impairment and in elderly patients

Aciclovir is eliminated by renal clearance; therefore the dose must be reduced in patients with renal impairment (see section Recommended Dosage). Elderly patients are likely to have reduced renal function and therefore the need for dose reduction must be considered in this group of patients. Both elderly patients and patients with renal impairment are at increased risk of developing neurological side effects and should be closely monitored for evidence of these effects. In the reported cases, these reactions were generally reversible on discontinuation of treatment (see section Adverse Reaction). The dosage of aciclovir should be adjusted in patients with severe renal impairment.

Aciclovir should be used cautiously in patients with underlying neurological abnormalities, significant hypoxia or serious renal, hepatic or electrolyte abnormalities, due to the potential encephalopathic changes.

The use of aciclovir in patients who have exhibited neurological reaction to cytotoxic drugs or are receiving concomitantly interferon or intrathecal methotrexate should be done with extra caution. Animal studies have shown that high doses aciclovir is cytotoxic.

Use in Pregnancy and Lactation

Few human data has shown that aciclovir can pass into the breast milk. Therefore aciclovir should only be prescribed to pregnant and lactating mothers unless the potential benefits outweigh the possibility of unknown risks.

Contraindications

Patients known to be hypersensitive to aciclovir and valaciclovir or any ingredients in these tablets.

Drug interaction

Aciclovir is eliminated primarily unchanged in the urine via active renal tubular secretion. Any drugs administered concurrently that compete with this mechanism may increase aciclovir plasma concentrations. Probenecid and cimetidine increase the AUC of aciclovir by this mechanism, and reduce aciclovir renal clearance.

Similarly increases in plasma AUCs of aciclovir and of the inactive metabolite of mycophenolate mofetil, an immunosuppressant agent used in transplant patients have been shown when the active substances are coadministered. However no dose adjustment is necessary because of the wide therapeutic index of aciclovir.

An experimental study on five male subjects indicates that concomitant therapy with aciclovir increases AUC of totally administered theophylline with approximately 50%. It is recommended to measure plasma concentrations during concomitant therapy with aciclovir.

Recommended Dosage

Herpes simplex infections

200mg aciclovir 5 times daily at 4 hours interval, daily for 5 days. Treatment may be extended to 10 days in severe initial infection. Treatment for Herpes simplex infection may be double to 400mg in severely immune compromised patients or in patients with impaired adsorption.

Prophylaxis of recurrent genital herpes simplex infection

- Immunologically healthy patients
200mg to 400mg aciclovir, four times daily (every six hours). Alternatively, 400mg aciclovir can be taken twice a day (every 12 hours). Effective prophylaxis can also be achieved with 200mg three times a day (every eight hours) or 200mg twice a day (every twelve hours).
- Immunosuppressed patients
200mg to 400mg aciclovir, four times daily (every six hours) can be used in severely immuno-compromised patients or in patients with impaired adsorption.

Varicella and Herpes zoster infections

Take 800mg aciclovir 5 times daily at 4hours intervals, omitting the night time dose. Treatment should continue for five to seven days.

Dosage in children

For treatment of Herpes simplex infections, and for prophylaxis of Herpes simplex infections in the immune-compromised, children aged two years and over should be given adult dosages and children below the age of two years should be given half the adult dose.

For treatment of Varicella infections, children over the age of six years can be given 800mg Acyclovir four times daily and children between the ages of two and six years can be given 400mg acyclovir four times daily.

Children below the age of two years can be given 200mg acyclovir four times daily. Dosing may be more accurately calculated as 20mg acyclovir/kg bodyweight (not to exceed 800mg) four times daily.

Treatment should continue for five days.

Dosage in the elderly

The possibility of renal impairment in the elderly must be considered and the dose should be adjusted accordingly (see Dosage in renal impairment below).

Adequate hydration of elderly patients taking high oral doses of aciclovir should be maintained.

Dosage in renal impairment

In the management of Herpes simplex infections in patients with severe renal impairment (creatinine clearance less than 10ml/minute) an adjustment of dosage to 200mg twice daily at approximately twelve-hourly intervals is recommended.

In the treatment of Herpes zoster infections it is recommended to adjust the dosage to 800mg twice daily, at approximately twelve-hourly intervals, for patients with severe renal impairment (creatinine clearance less than 10 ml/minute and to 800mg three times daily, at intervals of approximately eight hours, for patients with moderate renal impairment (creatinine clearance in the range 10 to 25ml/minute).

The tablet should be taken unchewed after meals with sufficient amount of liquid (a glass of water). In cases of impaired renal function, patients should be reminded to maintain hydration by taking sufficient water.

Symptoms and Treatment of Overdosage and antidote(s)

Aciclovir is only partly absorbed in the gastrointestinal tract. Patients have ingested overdoses of up to 20 g aciclovir on a single occasion, usually without toxic effects. Accidental, repeated overdoses of oral aciclovir over several days have been associated with gastrointestinal effects (such as nausea and vomiting) and neurological effects (headache and confusion).

Treatment

Patients should be observed closely for signs of toxicity. Haemodialysis significantly enhances the removal of aciclovir from the blood and may, therefore, be considered a management option in the event of symptomatic overdose.

Presentation:

ACIC 200 Tablet: Blister pack of 25 tablets and 100 tablets

ACIC 400 Tablet: Blister pack of 35 tablets and 70 tablets

ACIC 800 Tablet: Blister pack of 35 tablets

Storage conditions

Store in a cool, dry place **below 30°C.**

Protect from light.

Keep out of reach of children

Jauhi daripada kanak-kanak

Shelf life: ACIC 200/400/800 Tablet - Please refer to the outer carton.

Manufactured by:

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For:

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Revised date: 19 February 2021